Gal Arnon

Postdoctoral Researcher · Bocconi University

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Research Interests

I am broadly interested in cryptography and complexity theory. My research focuses on probabilistic proof systems (e.g., PCPs and IOPs) as well as their cryptographic counterparts, with an emphasis on succinct non-interactive arguments. I study both the theoretical foundations and practical aspects of these systems.

Appointments

2025–present **Postdoctoral Researcher, Bocconi University**
Milan, Italy. Hosted by Prof. Alon Rosen.

2025 **Research Fellow, Simons Institute for the Theory of Computing**
Berkeley, USA.

Education

2020–2025 **Ph.D. Computer Science**, *Weizmann Institute of Science*
Advisors: Prof. Moni Naor and Dr. Eylon Yogev.
Thesis: *New Advancements in Interactive Oracle Proofs: Theory, Practice, and Limitations*.

2017–2020 **M.Sc. Computer Science**, *Weizmann Institute of Science*
Advisor: Prof. Guy N. Rothblum.
Thesis: *On Prover-Efficient Public-Coin Emulation of Interactive Proofs*.

2013–2017 **B.Sc. Electrical Engineering & Computer Science**, *Tel Aviv University*
Magna Cum Laude.

Awards

2024 **Best Paper Award, Crypto 2024**
For "STIR: Reed–Solomon Proximity Testing with Fewer Queries".

2024 **Esther Hellinger Memorial Prize for Academic Excellence**, *Weizmann Institute of Science*

Papers

Preprints

4. **Complexity-Theoretic Consequences of Zero-Knowledge with Inefficient Simulation and Witness Indistinguishability**
   Gal Arnon, Noam Mazor, Rafael Pass, and Jad Silbak

3. **Witness Indistinguishable Arguments of Knowledge and One-Way Functions**
   Gal Arnon, Noam Mazor, Rafael Pass, and Jad Silbak

2. **Interactive Proofs for Batch Polynomial Evaluation**
   Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev

1. **Pairing-Based SNARGs with Two Group Elements**
   Gal Arnon, Jesko Dujmovic, and Eylon Yogev

Publications

11. **Designated-Verifier SNARGs with One Group Element**
    Gal Arnon, Jesko Dujmovic, and Yuval Ishai
    CRYPTO 2025

10. **Towards a White-Box Secure Fiat-Shamir Transformation**
    Gal Arnon and Eylon Yogev
    CRYPTO 2025, ZKProof 7.

9. **Instance Compression, Revisited**
   Gal Arnon, Shany Ben-David, and Eylon Yogev
   EUROCRYPT 2025

8. **WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification**
   Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
   EUROCRYPT 2025, ZKSummit 12, ZKProof 7.

7. **Hamming Weight Proofs of Proximity with One-Sided Error**
   Gal Arnon, Shany Ben-David, and Eylon Yogev
   TCC 2024

6. **STIR: Reed–Solomon Proximity Testing with Fewer Queries**
   Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
   CRYPTO 2024, ZKSummit 11, ZKProof 6. **[Best Paper Award]**

5. **IOPs with Inverse Polynomial Soundness Error**
   Gal Arnon, Alessandro Chiesa, and Eylon Yogev
   FOCS 2023, ITC 2024 Highlights

4. **A Toolbox for Barriers on Interactive Oracle Proofs**
   Gal Arnon, Amey Bhangale, Alessandro Chiesa, and Eylon Yogev

TCC 2022

3. **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**
   Gal Arnon, Alessandro Chiesa, and Eylon Yogev
   CCC 2022

2. **A PCP Theorem for Interactive Proofs and Applications**
   Gal Arnon, Alessandro Chiesa, and Eylon Yogev
   EUROCRYPT 2022

1. **On Prover-Efficient Public-Coin Emulation of Interactive Proofs**
   Gal Arnon and Guy N. Rothblum
   ITC 2021

Media Coverage

Quanta Magazine	*Computer Scientists Figure Out How to Prove Lies* mentions our work "Towards a White-Box Secure Fiat-Shamir Transformation".
ZK Podcast	Episode 321 - Interview on the STIR protocol.

Selected Talks

- **Fiat-Shamir Attacks and Mitigations**
  - Proofs Workshop, "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. *July 2025.*
  - Proofs Reading Group, "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. *June 2025.*
  - NTT Cryptography Seminar (Virtual). *April 2025.*
- **Instance Compression, Revisited**
  - Foundations Reading Group, "Cryptography, 10 Years Later" Semester, Simons Institute, Berkeley, USA. *July 2025.*
- **STIR: Reed–Solomon Proximity Testing with Fewer Queries**
  - CRYPTO 2024, Santa Barbara, USA. *August 2024.*
  - Interuniversity TCS Student Seminar, Tel Aviv University, Israel. *May 2024.*
  - Theory Lunch, Weizmann Institute of Science, Israel. *May 2024.*
  - HUJI TCS Seminar, Jerusalem, Israel. *May 2024.*
  - StarkWare Industries, Netanya, Israel. *April 2024.*
  - ZKSummit 11, Athens, Greece. *April 2024.*
- **IOPs with Inverse Polynomial Soundness Error**
  - ITC 2024 Highlights Track, Stanford University, USA. *August 2024.*
  - Technion TCS Seminar, Haifa, Israel. *February 2024.*
  - ZK Study Club (Virtual). *October 2023.*
  - StarkWare Industries, Netanya, Israel. *September 2023.*
  - Interuniversity TCS Student Seminar, Tel Aviv University, Israel. *July 2023.*
  - IST Austria TCS Seminar, Vienna, Austria. *June 2023.*
- **How To Be Convinced While Barely Listening (Even to Yourself)**
  - Theory Lunch, Weizmann Institute of Science, Israel. *May 2025.*

  - EPFL CS Theory Reading Group, Lausanne, Switzerland. *May 2023.*
  - Efficient Probabilistic Proofs Workshop, Bertinoro, Italy. *July 2022.*
- **A Toolbox for Barriers on Interactive Oracle Proofs**
  - TCC 2022, Chicago, USA. *November 2022.*
- **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**
  - CCC 2022, Philadelphia, USA (Virtual). *July 2022.*
- **A PCP Theorem for Interactive Proofs and Applications**
  - EUROCRYPT 2022, Trondheim, Norway. *May–June 2022.*
  - Theory Lunch, Weizmann Institute of Science, Israel. *July 2021.*
- **On Prover-Efficient Public-Coin Emulation of Interactive Proofs**
  - ITC 2021 (Virtual). *July 2021.*
  - "Proofs, Consensus, and Decentralizing Society" Program Seminar, Simons Institute, Berkeley, USA. *October 2019.*

Service

Proximity Prize	Committee member and judge for the Proximity Prize initiative of the Ethereum Foundation — a bounty to prove or disprove Reed–Solomon proximity gap conjectures.
Workshop	*Lattices Meet Hashes: Recent Advances in Post-Quantum Zero-Knowledge Proofs*. Postdoctoral workshop, Bernoulli Center, EPFL, Lausanne. Co-organized with Ngoc Khanh Nguyen. May 2023.
Program Comm.	TCC 2026
Reviews	CCC (2024) · CRYPTO (2019, 2022, 2023, 2024, 2025, 2026) · EUROCRYPT (2023, 2026) · FOCS (2025) · ITC (2025) · ITCS (2022, 2026) · SICOMP (2026) · SODA (2024) · STOC (2025) · TCC (2021, 2023, 2025)

Teaching

July 2023	**Teaching Assistant**, *Foundations and Frontiers of Probabilistic Proofs* MSRI (SLMath) Summer Graduate School, ETH Zurich, Switzerland.
July–Aug 2021	**Teaching Assistant**, *Foundations and Frontiers of Probabilistic Proofs* MSRI (SLMath) Summer Graduate School, Virtual.
Sept 2018	**Instructor**, *Mini-Course on Zero-Knowledge Proofs* Amos de-Shalit Summer School, Weizmann Institute of Science.